Jaime Hernandez-Vazquez

Computer Engineer

Contact

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Phone +1 (609)315-4974

Location

Region

Area Metropolitana de la Ciudad de Mexico

Professional Skills Summary

Game Development (Advanced proficiency): Unity Engine, C#, Godot, Unreal Engine, C++, GDScript, three.js

Web Development (Intermediate proficiency): HTML, CSS, Javascript, Svelte, Python, GCP, Oracle

Embedded Development (Basic proficiency): ESP-32, Raspberry PI, C/C++, ASM, Arduino, Digital design

Digital Artist (Hobbyist): 3D Modeling/Materials, Aseprite, Blender, Image Processing, Gen AI, Video Editing

Work Experience

2023-02 to Present

Computer Engineer:

Main software engineer in charge of design and develop a Mixed Reality Drivers' Education suite for Meta Quest devices. Implemented custom input device to create an immersive driving environment in any car. Academic research colaboration with UPenn XLab, published a couple of tech reports.

- Impemented custom sensors as input devices. Refactored firmware in C++ for Arduino using PlatformIO.
- Designed and implemented Companion app and custom hardware driver for sensor calibration using C# and Unity.
- Created Mixed Reality prototype for driving simulation using Unity, C# and several plugins.
- Research and development for Driving Simulation and Autonomous Vehicles focusing on young people using XR and custom hardware.
- Explored multiple Game Engine technologies for Drivers Education, Traffic Simulation and Urban Planning.

2021-09 to 2022-02

AR/VR Developer:

Main developer for Virtual Reality ArchViz and Special Ed apps using Unity. Created multople Instagram AR effects/filters using SparkAR Studio.

- Designed and prototyped a Unity-based VR educational app which pairs with a mobile companion app for remote control and monitor.
- Creative ArchViz interactive VR galleries for standalone Meta Quest devices.
- B2B ad filters and effects for Instagram and Facebook.

2018-06 to 2019-10

Sofware Developer:

Lead software developer in charge of maintenance and enhacement of RA Infinitum and Claro RA Unity augmented reality apps. Close colaboration with multiple animation, concept, illustration and 3D artists for high quality content creation and eye-candy detailing. Prototyping applications in Unity and developed multiple Meta Spark AR interactive effects.

- Optimized art team workflows for delivering PBR-ready content for AR apps and effects 4x faster.
- Reduced cloud services costs replacing Vuforia SDK with MaxstAR SDK. This improved customer support from server owners.
- Developed Spark AR Studio templates for further creation and release of dozens of Facebook and Instagram camera effects.
- Continuous Development and Deployment of base AR cloud application and their different skins using Unity and C#.
- Implemented a few games and interactive ads webpages.

2021-04 to 2021-08

Junior Front-end Engineer:

Improved and maintained in-house front-end applications while also aiding other teams in automating scripts for Google Workspace and data processing.

- Angular Framework
- Google Apps Script
- Tailwind CSS
- Svelte
- Typescript

Education

2008-08-15 to 2013-06-01

Engineering Faculty, Universidad Nacional Autonoma de México (UNAM).

- Computer Engineering: *Computer Graphics*

Languages

Spanish	Native speaker
English	Full proficency
German	B1 (ÖSD-ZD)
Chinese	Beginner

Personal

Generative AI, LLMs, Agents, Agentic development, Harness

C++, C#, Java, Design Patterns, Refactoring, SOLID

Databases, Networking, Clustering, Cloud, Rust, Go